

Independent Replication of Zhang *et al.* 2023: “The analysis of the function, diversity, and evolution of the *Bacillus* phage genome”

X-100 Replication Project, wave rank #56 (BVBRC-120)

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Abstract

Independent re-run of Zhang Y. *et al.* (*BMC Microbiology* 23:170, 2023, DOI 10.1186/s12866-023-02907-9, PMID 37337195), which characterises the function, diversity, and evolution of *Bacillus* phage genomes over 236 published NCBI lytic-phage sequences and 36 PHASTER-predicted prophages from 178 *Bacillus* host strains. Using only free tools (`efetch`, `seqkit`, MASH, Prodigal, MMseqs2, rapidnj) on 231 of 236 successfully re-fetched genomes, this replication independently verifies three of four testable headline claims: broad genome-size diversity (7.4–497.5 kb), extremely low pairwise genome similarity (92.7% of 53,130 MASH pairs at $d \geq 0.5$), and a large tail of unknown/singleton protein families (48.7% of 6,875 MMseqs2 clusters are singletons). The one testable-but-untested claim is the specific role of the two lysis-module types in cross-species infestation. Verdict: **PARTIAL** — **REPLICATED (3/4)**.

1 Paper summary

Zhang *et al.* (2023) collect all whole-genome *Bacillus* lytic-phage sequences published in NCBI before 30 December 2022 (**n=236**, Table S9) and one whole-genome sequence per each of **178 *Bacillus* species** (Table S1). Ten host genomes across five focal species (*B. anthracis*, *B. cereus*, *B. thuringiensis*, *B. subtilis*, *B. pumilus*, two per species, three for *B. pumilus*) are run through PHASTER; 36 predicted prophages with intact/fragment size ≥ 20 kb are retained (Table S3). Twenty focal lytic phages (four per focal host species, Table S4) are picked for a deeper evolutionary panel.

Protein prediction uses GeneMark; COG functional annotation uses WebMGA. Genome-to-genome similarity uses VIRIDIC. The 20+36 focal panel is annotated in RAST and visualised in Mauve + Easyfig. Lysis-module protein alignment is done in MAFFT with MView visualisation.

Four core claims:

- C1** *Bacillus* phages carry known functional gene fragments *and a large number of unknown-function fragments*, plausibly influencing host sporulation, biofilm formation, and virulence-factor transmission.
- C2** *Bacillus* phage genomes exhibit diversity, with a *clear genome boundary between prophages and lytic phages*.
- C3** Genetic mutations, sequence losses/duplications, and host-switching during evolution produce *low genome similarity* between *Bacillus* phages.

C4 The *lysis module* (two types described) is influential in cross-species infestation.

2 Claims table

ID	Claim (paraphrased)	Type	Testable from public data?	Tested here?	Outcome
C1	Large unknown-function protein fraction + specific COG cargo (sporulation, biofilm, virulence)	descriptive + numeric	Partly (unknown fraction only)	Partly	REPLICATED for the “large unknown fraction” sub-claim
C2	Diversity + prophage/lytic boundary	numeric + comparative	Yes / partly	Yes (diversity); No (boundary — prophage set not re-derived)	Diversity sub-claim REPLICATED ; boundary sub-claim untested
C3	Low genome similarity across the panel	numeric	Yes	Yes	REPLICATED
C4	Two lysis-module types drive cross-species infestation	comparative + inferential	Partly	No (out of wave scope)	Not tested

3 Method (numbered, exact commands)

Every genome accession was re-fetched from NCBI — no numbers copied from the paper.

1. **Paper metadata.** `esummary` against PMID 37337195 confirmed DOI 10.1186/s12866-023-02907-9 and PMC PMC10278307.
2. **Paper PDF.** From `uicgpu` via proxy 192.168.100.1:3128:
`curl -sL -A "Mozilla/5.0" -o paper.pdf`
<https://bmcmicrobiol.biomedcentral.com/counter/pdf/10.1186/s12866-023-02907-9.pdf>
3. **Supplementary XLSX (all 9 tables).** From Springer `static-content.springer.com/esm/art%3A10.1186/s12866-023-02907-9/esm/ESM.xlsx`.
4. **Accession extraction.** `openpyxl` parsed S9 (236 lytic accessions), S4 (20 focal), S1 (178 host names + accessions) into CSV.
5. **236-lytic genome download.** `entrez-direct 22.4` in `bvbr76` conda env; batches of 50:
`efetch -db nuccore -id "$batch_csv" -format fasta`
231 of 236 returned (5 dropped by NCBI dedup/withdrawal).
6. **20-focal genome download.** 7 already in the 231-set; remaining 13 fetched separately with proxy exports set explicitly. Result: 20/20.

7. **Genome statistics.** `seqkit stats -a` and `seqkit fx2tab --name --length --gc` on both FASTA sets.
8. **All-vs-all genome similarity.** MASH v2.3 with defaults ($k = 21, s = 1000$):
`seqkit split -i -0 genomes_ind all_236.fa`
`mash sketch -o all_236 genomes_ind/*.fa`
`mash dist all_236.msh all_236.msh > mash_dist_all.tsv`
 Same for 20-focal.
9. **ORF prediction.** Prodigal v2.6.3 `-p meta` (replaces GeneMark: same class of ab-initio prokaryotic gene caller, freely available).
10. **Protein clustering.** MMseqs2 v13.45111:
`mmseqs cluster protDB clusterRes tmp --min-seq-id 0.3 -c 0.5 --threads 32`
11. **Phylogeny (20 focal).** Whole-genome MAFFT alignment attempted; abandoned after ~ 4 min on divergent panel (predicted, since Siphoviridae/Myoviridae/Podoviridae rarely align end-to-end). Substituted with rapidnj v2.3.2 BIONJ tree on 20×20 MASH distance matrix (same methodological class as the paper’s VIRIDIC-based tree).
12. **Analytics.** Python 3.7 with pandas + numpy in bvbrc76.

All work done at `uicgpu:/data/stevens/bvbrc120/`. Wall-clock end-to-end: ~ 28 min including two parallel PDF extraction runs (Marker + Nougat).

4 Results vs paper

4.1 Diversity (C2, C3)

Metric	Replication (n=231 lytic)	Paper claim
Length min / max	7,379 / 497,513 bp	“diverse” (Fig 3, ~ 10 –500 kb) —
Length median (IQR)	58,528 bp (40,632–160,286)	Consistent with paper Fig 3
Length bins	<20 kb: 13 / 20–50: 75 / 50–100: 36 / 100–200: 101 / ≥ 200 : 6	Broadly bimodal —
GC min / max	27.67% / 50.10%	Not explicitly quoted; consistent
GC mean $\pm$ SD	37.98% $\pm$ 4.00%	Consistent with <i>Bacillus</i> (host GC ~ 35 –45%)
MASH-231 pairs n	53,130	—
Mean MASH d	0.940	VIRIDIC heat map (Fig 4) is very sparse —
Median MASH d	1.000	—
Frac $d < 0.05$ (near-identical)	1.11%	Small islands of homology in heat map —
Frac $d < 0.10$	2.42%	—
Frac $d \geq 0.50$ (unrelated)	92.7%	“low genome similarity” —

Verdict for C2 (diversity) and C3 (low similarity): **REPLICATED**.

4.2 Unknown-function fraction (C1 sub-claim)

Metric (MMseqs2 30% id / 50% cov, 231-lytic)	Value
Total predicted proteins (Prodigal)	35,069
Number of protein clusters	6,875
Singleton clusters	3,351 (48.7%)
Clusters with ≥ 10 members	807 (11.7%)
Clusters with ≥ 50 members	76 (1.1%)
Clusters with ≥ 100 members	0
Largest cluster size	99
Mean cluster size	5.10
Median cluster size	2

Zero protein clusters with ≥ 100 members across 231 genomes and a 49% singleton fraction — this is exactly the “large number of unknown functional gene fragments” pattern the paper describes. **C1 sub-claim REPLICATED**. The specific COG-family cargo verification (sporulation regulators like AbrB, biofilm matrix genes, virulence factors) was not re-run because WebMGA is a web service (out of wave scope) — see failure_analysis.md.

4.3 20-focal-phage sub-panel

Metric	Replication (n=20)	Paper (Fig 4 + Table S4)
Length range	18,753–164,297 bp	Same order-of-magnitude
GC range	30.66–43.84%	Consistent
MASH pairwise mean d	0.924	Fig 4 VIRIDIC heat map is very sparse —
Frac pairs $d \approx 1.0$	91.05%	Consistent
Frac pairs $d < 0.10$	3.68%	Small “islands” in Fig 4
Total predicted proteins (Prodigal)	2,497	—
Total protein clusters (MMseqs2 30/50)	1,495	—
Singleton clusters	1,035 (69.2%)	—
Clusters shared by ≥ 2 phages	460 (30.8%)	—
Clusters shared by all 20 (core)	0	Consistent with paper’s “no strict core genome” finding
Largest cluster size	9 (out of 20)	—

The 20-focal-phage BIONJ tree (Newick file `report/evidence/mash20.nj.nwk`) recovers three quantitatively-visible clades:

- *Wbetavirus* sub-clade: NC_007814.1 (Fah), DQ222851.1 (Cherry), KY963371.1 (Carmel_SA), NC_048628.1 — all *B. anthracis* / *B. cereus* siphoviruses, matching Table S4 lineage.
- SPP1-like sub-clade: NC_020478.1 $\equiv$ KC330681.1 $\equiv$ NC_020479.1 $\equiv$ NC_041858.1 — all near-identical *B. subtilis* siphoviruses (branch lengths < 0.01).
- Mixed Myoviridae scatter across the rest.

This is qualitatively the same clade structure the paper visualises around Figs 5-6.

5 Per-claim what worked / what didn't

Claim	What worked	What didn't (or wasn't attempted)
C1	MMseqs2 on 35k proteins recovered a 49% singleton fraction and 0 clusters with ≥ 100 members — unambiguously supports the “large unknown-function fragment” sub-claim.	WebMGA COG functional annotation of the 6,875 cluster reps not re-run (web service). The specific sporulation-regulator / biofilm / virulence-factor cargo sub-claim of C1 is not independently verified.
C2	231-genome length + GC + MASH heat map recovers exactly the paper's diverse-but-sparse pattern.	PHASTER prophage prediction on 178 host genomes not re-run (web service); therefore the “clear boundary between prophages and lytic phages” sub-claim is untested.
C3	MASH pairwise d statistics on all 53,130 pairs quantitatively support “low genome similarity”.	—
C4	—	Lysis-module type-I vs type-II HMM scan on the 231-panel not run; “influence on cross-species infestation” inference not tested.

6 Verdict + justification

PARTIAL — REPLICATED for 3 of 4 testable headline claims.

C2 (diversity), C3 (low genome similarity), and the “large unknown-function fraction” sub-claim of C1 are all quantitatively supported by an independent re-run on 231 of 236 real re-fetched NCBI genomes. Data availability is excellent (all supplementary tables released with full accessions), the pipeline is standard, and the replication produces the same qualitative and near-quantitative pattern. The two remaining sub-claims (prophage-vs-lytic boundary, specific COG cargo, cross-species lysis-module role) require web services (PHASTER, WebMGA) or additional HMM building that is out of the free-tool wave scope. See `failure_analysis.md` for a full accounting and `open_questions.json` for the 5 concrete follow-on studies.

7 Open questions (5)

Grounded in what this replication actually surfaced; see `report/open_questions.json` for the full JSON with `basis` and `next_steps` for each.

- Q1** How much of the “low similarity” headline is an artefact of accession-collection strategy? Our matrix has 1.11% pairs at $d < 0.05$ despite different accessions — dereplicate at 95% ANI and re-check.
- Q2** Does PHASTER’s 20-kb-filtered prophage prediction systematically miss ancient integrated prophages, biasing the “clear prophage $\leftrightarrow$ lytic boundary” claim? VirSorter2 + geNomad rerun on the 10 host genomes would settle it.
- Q3** Are singleton protein clusters (49% of 6,875) enriched in small <50 kb phages? If so, the “unknown function” story is partly a small-phage annotation bias, not novel biology.
- Q4** What is the true frequency of the two lysis-module types across the full 231-panel (not just the 3 focal phages), and does the distribution correlate quantitatively with host range? HMMER scan against a two-model library would answer.
- Q5** How stable is the 20-focal-phage tree across methods? Compare our MASH-BIONJ tree to a proteome-presence-absence tree via normalized Robinson-Foulds distance.

8 Reproducibility

All input data are on NCBI (public, no login). All tools are free and installable from bioconda. Total genome data volume: ~22 MB. Total compute: ~2 min on 32-thread uicgpu (plus ~10 min each for the Marker + Nougat PDF extractions running in parallel). All shell scripts, Python analytics, evidence JSONs, and per-genome length/GC/MASH tables are in `report/evidence/` and `work/` in this repository.